Scattering Amplitudes: From QFT to Modern Methods

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Part I

Quantum Field Theory (QFT Foundations)

Chapter 1

Quantum Field Theory (QFT) Foundations

1.1 LSZ Reduction Formula

1.2 Path Integrals

1.3 Feynman Rules for Gauge Theories

1.4 Non-Abelian Gauge Theory

Chapter 2

Helicity States and Weyl Spinors

This chapter builds the spinor algebra that underlies everything in Parts II and III. The physical target is simple: massless particles carry definite *helicity* $h = \pm\frac{1}{2}$ (or ± 1 for gluons), and the two-component Weyl spinors λ_α and $\tilde{\lambda}_{\dot{\alpha}}$ are precisely the objects that represent those helicity eigenstates. We establish this connection carefully here so that Chapter 3 can use the spinor-bracket formalism without re-deriving its foundations.

2.1 Lorentz Group Representations

The Lorentz algebra $\mathfrak{so}(1,3)$ is (complexified) isomorphic to $\mathfrak{su}(2)_L \oplus \mathfrak{su}(2)_R$ via the non-Hermitian combinations

$$N_i \equiv \tfrac{1}{2}(J_i - iK_i), \qquad N_i^\dagger \equiv \tfrac{1}{2}(J_i + iK_i), \tag{2.1}$$

where J_i are rotation generators and K_i are boost generators. They satisfy $[N_i, N_j] = i\varepsilon_{ijk}N_k$, $[N_i^\dagger, N_j^\dagger] = i\varepsilon_{ijk}N_k^\dagger$, $[N_i, N_j^\dagger] = 0$, so irreducible representations are labelled by two half-integers $(n,\, n')$.

(n, n')	Srednicki	Dim	Field	Index type
$(0,0)$	$(1,1)$	1	scalar ϕ	none
$(\frac{1}{2},0)$	$(2,1)$	2	left-handed Weyl ψ_α	undotted
$(0,\frac{1}{2})$	$(1,2)$	2	right-handed Weyl $\tilde{\psi}_{\dot{\alpha}}$	dotted
$(\frac{1}{2},\frac{1}{2})$	$(2,2)$	4	vector A^μ	spacetime

The key point for helicity: under a rotation by angle θ about the momentum axis (the little group of a massless particle), a $(\frac{1}{2},0)$ field picks up phase $e^{-i\theta/2}$ (helicity $-\frac{1}{2}$) and a $(0,\frac{1}{2})$ field picks up $e^{+i\theta/2}$ (helicity $+\frac{1}{2}$).

2.2 Left-Handed Weyl Spinors ψ_α

A left-handed Weyl field $\psi_\alpha(x)$ lives in the $(\frac{1}{2},0)$ representation. Under an infinitesimal Lorentz transformation $\delta\omega_{\mu\nu}$:

$$\delta\psi_\alpha = \tfrac{i}{2}\,\delta\omega_{\mu\nu}\,(S_{\mathrm{L}}^{\mu\nu})_\alpha{}^\beta\,\psi_\beta. \tag{2.2}$$

The six generators are explicitly (Srednicki eqs. 34.9–34.10):

$$(S_{\mathrm{L}}^{ij})_\alpha{}^\beta = \tfrac{1}{2}\varepsilon^{ijk}\sigma_k, \qquad (S_{\mathrm{L}}^{k0})_\alpha{}^\beta = \tfrac{i}{2}\sigma_k, \tag{2.3}$$

where σ_k are the Pauli matrices:

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{2.4}$$

These satisfy the Lorentz algebra $[S_L^{\mu\nu}, S_L^{\rho\sigma}] = i(g^{\nu\rho} S_L^{\mu\sigma} - \cdots)$. The boost generators S_L^{k0} are anti-Hermitian (the Lorentz group is non-compact), reflecting that boosts are not unitary.

2.3 Right-Handed Weyl Spinors $\tilde{\psi}_{\dot{\alpha}}$

Hermitian conjugation swaps $N_i \leftrightarrow N_i^\dagger$, sending $(\frac{1}{2}, 0) \to (0, \frac{1}{2})$. So

$$(\psi_\alpha)^\dagger \equiv \tilde{\psi}_{\dot{\alpha}}(x) \tag{2.5}$$

is a *right-handed* Weyl field with a *dotted* index. Its generators are (Srednicki eq. 34.17):

$$\boxed{(S_{\mathrm{R}}^{\mu\nu})_{\dot{\alpha}}{}^{\dot{\beta}} = -[(S_{\mathrm{L}}^{\mu\nu})_\alpha{}^\beta]^*} \tag{2.6}$$

Concretely this reverses the sign of boosts (consistent with $L \leftrightarrow R$ under parity) and from eq. (35.22):

$$S_{\mathrm{R}}^{\mu\nu} = -\tfrac{i}{4}(\bar{\sigma}^\mu \sigma^\nu - \bar{\sigma}^\nu \sigma^\mu). \tag{2.7}$$

Convention note. We adopt Srednicki's metric $g^{\mu\nu} = \mathrm{diag}(+1, -1, -1, -1)$ and $\varepsilon^{12} = \varepsilon_{21} = +1$ throughout. The amplitudes literature often uses the same conventions; we flag any deviation when it appears.

2.4 The ε Symbol: SL$(2, \mathbb{C})$ Metric

From $(\frac{1}{2}, 0) \otimes (\frac{1}{2}, 0) = (0, 0)_A \oplus (1, 0)_S$ there is a unique (up to scale) invariant antisymmetric symbol. With Srednicki's normalization (eq. 34.22 / 35.1):

$$\varepsilon^{12} = \varepsilon_{21} = +1, \qquad \varepsilon^{21} = \varepsilon_{12} = -1, \tag{2.8}$$

so in matrix form:

$$\varepsilon_{\alpha\beta} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \varepsilon^{\alpha\beta} = \begin{pmatrix} 0 & +1 \\ -1 & 0 \end{pmatrix}, \qquad \varepsilon_{\alpha\beta}\varepsilon^{\beta\gamma} = \delta_\alpha{}^\gamma. \tag{2.9}$$

The same structure holds for dotted indices.

2.4.1 Raising and lowering

$$\psi^\alpha = \varepsilon^{\alpha\beta}\psi_\beta, \qquad \psi_\alpha = \varepsilon_{\alpha\beta}\psi^\beta. \tag{2.10}$$

Sign trap (Srednicki eq. 34.27):

$$\psi^a \chi_a = \varepsilon^{ab}\psi_b\chi_a = -\varepsilon^{ba}\psi_b\chi_a = -\psi_b\chi^b. \tag{2.11}$$

Hence $\psi^\alpha\chi_\alpha = -\psi_\alpha\chi^\alpha$: *index position matters and carries a sign.*

2.5 Lorentz-Invariant Spinor Products

The fundamental Lorentz-invariant bilinears are:

$$\psi\chi \equiv \psi^\alpha \chi_\alpha = \varepsilon^{\alpha\beta}\psi_\alpha\chi_\beta, \tag{2.12}$$

$$\tilde{\psi}\tilde{\chi} \equiv \tilde{\psi}_{\dot\alpha}\tilde{\chi}^{\dot\alpha} = \varepsilon_{\dot\alpha\dot\beta}\tilde{\psi}^{\dot\alpha}\tilde{\chi}^{\dot\beta}. \tag{2.13}$$

For Grassmann-valued fields, $\psi\chi = +\chi\psi$ (two minus signs cancel; Srednicki eq. 35.25), and $(\psi\chi)^\dagger = \tilde{\chi}\tilde{\psi}$.

These are the precursors of the spinor brackets $\langle ij\rangle$ and $[ij]$ introduced in Chapter 3: when $\psi_\alpha = \lambda_{i\alpha}$ and $\chi_\alpha = \lambda_{j\alpha}$ are the spinors of massless momenta p_i, p_j, then $\langle ij\rangle = \varepsilon^{\alpha\beta}\lambda_{i\alpha}\lambda_{j\beta}$ and $[ij] = \varepsilon_{\dot\alpha\dot\beta}\tilde{\lambda}_i^{\dot\alpha}\tilde{\lambda}_j^{\dot\beta}$.

2.6 The σ^μ Symbol: Vector–Spinor Dictionary

A field in the $(\frac{1}{2}, \frac{1}{2})$ representation carries one undotted and one dotted index. The dictionary to 4-vectors is (Srednicki eq. 34.28–34.30):

$$A_{\alpha\dot\alpha} = \sigma^\mu_{\alpha\dot\alpha} A_\mu, \qquad \sigma^\mu_{\alpha\dot\alpha} = (I, \vec{\sigma}). \tag{2.14}$$

Explicitly:

$$\sigma^0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \ \sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \ \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \ \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{2.15}$$

Raising both spinor indices via the ε symbols gives (Srednicki eq. 35.19–35.20):

$$\bar{\sigma}^{\mu\,\dot\alpha\alpha} \equiv \varepsilon^{\alpha\beta}\varepsilon^{\dot\alpha\dot\beta}\,\sigma^\mu_{\beta\dot\beta} = (I, -\vec{\sigma}). \tag{2.16}$$

2.6.1 Key identities

$$\sigma^\mu_{\alpha\dot\alpha}\,\sigma_{\mu\beta\dot\beta} = -2\,\varepsilon_{\alpha\beta}\,\varepsilon_{\dot\alpha\dot\beta}, \tag{35.4}$$

$$\varepsilon^{\alpha\beta}\varepsilon^{\dot\alpha\dot\beta}\,\sigma^\mu_{\alpha\dot\alpha}\,\sigma^\nu_{\beta\dot\beta} = -2\,g^{\mu\nu}, \tag{35.5}$$

$$\mathrm{tr}(\sigma^\mu\bar{\sigma}^\nu) \equiv \sigma^\mu_{\alpha\dot\alpha}\,\bar{\sigma}^{\nu\,\dot\alpha\alpha} = 2\,g^{\mu\nu}, \tag{2.17}$$

$$\sigma^\mu\bar{\sigma}^\nu + \sigma^\nu\bar{\sigma}^\mu = -2\,g^{\mu\nu}\,I_2. \tag{35-Clifford}$$

The last line is the Weyl-spinor analogue of the Dirac Clifford algebra. The generators are its antisymmetric part:

$$S_{\mathrm{L}}^{\mu\nu} = \tfrac{i}{4}(\sigma^\mu\bar{\sigma}^\nu - \sigma^\nu\bar{\sigma}^\mu). \tag{35.21}$$

2.6.2 Momentum bi-spinor

For any 4-momentum p^μ, define

$$p_{\alpha\dot\alpha} \equiv \sigma^\mu_{\alpha\dot\alpha}\,p_\mu = \begin{pmatrix} p^0 + p^3 & p^1 + ip^2 \\ p^1 - ip^2 & p^0 - p^3 \end{pmatrix}. \tag{2.18}$$

Then $\det(p_{\alpha\dot\alpha}) = (p^0)^2 - |\mathbf{p}|^2 = p^2$, so $p^2 = 0$ iff the matrix has rank 1 — the factorization $p_{\alpha\dot\alpha} = \lambda_\alpha\tilde{\lambda}_{\dot\alpha}$ is equivalent to masslessness. This is the starting point of Chapter 3.

2.7 Helicity from the Little Group

For a massless particle with momentum p^μ, the little group (the subgroup of the Lorentz group that fixes p^μ) is $ISO(2)$, whose compact part is $U(1)$. Under the $U(1)$ rotation by angle θ, a field of helicity h transforms as $e^{-2ih\theta}$.

In terms of spinors: the little-group generator rescales

$$\lambda_\alpha \to e^{i\theta/2}\lambda_\alpha, \qquad \tilde{\lambda}_{\dot\alpha} \to e^{-i\theta/2}\tilde{\lambda}_{\dot\alpha}, \tag{2.19}$$

i.e. the continuous parameter $t = e^{i\theta/2}$ acts as

$$\lambda \to t\lambda, \qquad \tilde{\lambda} \to t^{-1}\tilde{\lambda}. \tag{2.20}$$

A state created by λ_α has helicity $-\frac{1}{2}$ (left-handed, positive-chirality), and one created by $\tilde{\lambda}_{\dot\alpha}$ has helicity $+\frac{1}{2}$ (right-handed, negative-chirality).

For a spin-1 gluon the helicity-± 1 polarization vectors scale as $\varepsilon^\pm_\mu \to t^{\mp 2}\varepsilon^\pm_\mu$ under (2.20); this determines the form of $\varepsilon^\pm$ in terms of spinors (see §3.5).

2.8 The Weyl Lagrangian

For a single left-handed Weyl field ψ_α, the Lorentz-invariant hermitian Lagrangian is (Srednicki eq. 36.2):

$$\boxed{\mathcal{L} = i\tilde{\psi}_{\dot\alpha}\,\bar{\sigma}^{\mu\,\dot\alpha\alpha}\,\partial_\mu\psi_\alpha - \tfrac{1}{2}m\,\varepsilon^{\alpha\beta}\psi_\alpha\psi_\beta - \tfrac{1}{2}m\,\varepsilon_{\dot\alpha\dot\beta}\tilde{\psi}^{\dot\alpha}\tilde{\psi}^{\dot\beta}.} \tag{2.21}$$

The kinetic term is real modulo a total divergence. The mass term is a *Majorana mass*: it breaks the $U(1)$ symmetry $\psi \to e^{i\alpha}\psi$ and is non-zero because ψ_α are Grassmann-valued.

2.8.1 Equations of motion

Varying with respect to $\tilde{\psi}^{\dot\alpha}$:

$$-i\,\bar{\sigma}^{\mu\,\dot\alpha\alpha}\,\partial_\mu\psi_\alpha + m\,\tilde{\psi}^{\dot\alpha} = 0. \tag{36.4}$$

For $m = 0$ this is the **Weyl equation** for a massless left-handed spinor. It is the field equation whose plane-wave solutions give the spinors $\lambda_\alpha(p)$ used in amplitude calculations.

2.8.2 Gamma matrices: Weyl representation

Assembling ψ_α and $\tilde{\psi}_{\dot\alpha}$ into a 4-component Majorana spinor $\Psi = (\psi_\alpha, \tilde{\psi}^{\dot\alpha})^T$, the equations of motion take the Dirac form $(-i\gamma^\mu\partial_\mu + m)\Psi = 0$ with (Srednicki eq. 36.7):

$$\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu_{\alpha\dot\beta} \\ \bar{\sigma}^{\mu\,\dot\alpha\beta} & 0 \end{pmatrix}, \qquad \{\gamma^\mu, \gamma^\nu\} = -2g^{\mu\nu}I_4. \tag{36.7, 36.9}$$

The chiral projectors are $P_L = \frac{1}{2}(I - \gamma^5) = \mathrm{diag}(I_2, 0)$ and $P_R = \frac{1}{2}(I + \gamma^5) = \mathrm{diag}(0, I_2)$ with $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3 = \mathrm{diag}(-I_2, +I_2)$.

2.9 Majorana and Dirac Fermions

Property	Majorana	Dirac
Weyl fields	1 (ψ)	2 (χ, ξ)
$U(1)$ symmetry	No	Yes
Conserved charge	None	$Q = \int j^0 \, d^3x$
Self-conjugate ($\Psi^C = \Psi$)	Yes	No
Lagrangian	$\frac{i}{2}\bar{\Psi}\gamma^\mu\partial_\mu\Psi - \frac{m}{2}\bar{\Psi}\Psi$	$i\bar{\Psi}\gamma^\mu\partial_\mu\Psi - m\bar{\Psi}\Psi$
Real d.o.f.	2	4

In amplitude calculations with external gluons (massless spin-1), the internal spinors are always effectively massless Weyl spinors; the Majorana/Dirac distinction matters for external fermion lines.

2.10 Summary: Spinor Toolkit

Object	Expression / Rule
Left-handed field	ψ_α (undotted index, helicity $-\frac{1}{2}$)
Right-handed field	$\tilde{\psi}_{\dot{\alpha}} = (\psi_\alpha)^\dagger$ (dotted, $+\frac{1}{2}$)
Rotation generator	$(S_L^{ij})_\alpha{}^\beta = \frac{1}{2}\varepsilon^{ijk}\sigma_k$
Boost generator	$(S_L^{k0})_\alpha{}^\beta = \frac{i}{2}\sigma_k$
R-generators	$S_R^{\mu\nu} = -[S_L^{\mu\nu}]^*$
ε normalization	$\varepsilon^{12} = \varepsilon_{21} = +1$
Raise/lower	$\psi^\alpha = \varepsilon^{\alpha\beta}\psi_\beta, \quad \psi_\alpha = \varepsilon_{\alpha\beta}\psi^\beta$
Sign identity	$\psi^\alpha\chi_\alpha = -\psi_\alpha\chi^\alpha$
Angle bracket	$\psi\chi = \varepsilon^{\alpha\beta}\psi_\alpha\chi_\beta = +\chi\psi$
Square bracket	$\tilde{\psi}\tilde{\chi} = \varepsilon_{\dot{\alpha}\dot{\beta}}\tilde{\psi}^{\dot{\alpha}}\tilde{\chi}^{\dot{\beta}}$
σ^μ	$(I, \vec{\sigma}), \quad \bar{\sigma}^\mu = (I, -\vec{\sigma})$
Trace	$\mathrm{tr}(\sigma^\mu\bar{\sigma}^\nu) = 2g^{\mu\nu}$
Clifford	$\sigma^\mu\bar{\sigma}^\nu + \sigma^\nu\bar{\sigma}^\mu = -2g^{\mu\nu}I_2$
Momentum bi-spinor	$p_{\alpha\dot{\alpha}} = \sigma^\mu_{\alpha\dot{\alpha}}p_\mu, \quad \det = p^2$
Little group	$\lambda \to t\lambda$, $\tilde{\lambda} \to t^{-1}\tilde{\lambda}$
Weyl Lagrangian	$i\tilde{\psi}\bar{\sigma}^\mu\partial_\mu\psi - \frac{m}{2}\psi\psi - \frac{m}{2}\tilde{\psi}\tilde{\psi}$
Weyl equation ($m = 0$)	$i\bar{\sigma}^{\mu\,\dot{\alpha}\alpha}\partial_\mu\psi_\alpha = 0$

Next: Chapter 3 uses the momentum bi-spinor factorization $p_{\alpha\dot{\alpha}} = \lambda_\alpha\tilde{\lambda}_{\dot{\alpha}}$ (which follows directly from $p^2 = 0$ and the rank-1 condition above) to build the spinor-bracket formalism and polarization vectors for massless gauge bosons.

Part II

Spinor-Helicity Formalism

Chapter 3

Spinor-Helicity Formalism

3.1 The Factorization of Null Momenta

For massless particles, the momentum p^μ satisfies $p^2 = 0$. In the Van der Waerden notation, the momentum bi-spinor is defined via the Pauli matrices $\sigma^\mu = (\mathbf{1}, \vec{\sigma}_{\text{Pauli}})$:

$$p_{\alpha\dot{\alpha}} = p^\mu (\sigma_\mu)_{\alpha\dot{\alpha}}. \tag{3.1}$$

Explicitly in 2×2 matrix form:

$$p_{\alpha\dot{\alpha}} = \begin{pmatrix} p^0 + p^3 & p^1 + ip^2 \\ p^1 - ip^2 & p^0 - p^3 \end{pmatrix}. \tag{3.2}$$

For massless p ($p^2 = 0$), $\det(p_{\alpha\dot{\alpha}}) = (p^0)^2 - |\mathbf{p}|^2 = 0$, so this matrix has rank 1 and factorizes as:

$$p_{\alpha\dot{\alpha}} = \lambda_\alpha \tilde{\lambda}_{\dot{\alpha}} \tag{3.3}$$

where λ_α and $\tilde{\lambda}_{\dot{\alpha}}$ are commuting (bosonic) Weyl spinors (Srednicki Sec. 20.3, 42.1). This factorization is unique up to the little group scaling:

$$\lambda_\alpha \to t\lambda_\alpha, \quad \tilde{\lambda}_{\dot{\alpha}} \to t^{-1}\tilde{\lambda}_{\dot{\alpha}}, \quad t \in \mathbb{C}^*. \tag{3.4}$$

For real momenta $t \in U(1)$; for complex momenta λ and $\tilde{\lambda}$ are independent. A left-handed (positive helicity $+\frac{1}{2}$) Weyl spinor scales as $\lambda \to t\lambda$; a right-handed one as $\tilde{\lambda} \to t^{-1}\tilde{\lambda}$.

3.2 Spinor Brackets

The fundamental Lorentz-invariant contractions are (Srednicki eq. 42.16):

$$\langle ij \rangle = \langle ij \rangle = \epsilon^{\alpha\beta} \lambda_{i\alpha} \lambda_{j\beta} = \lambda_i^0 \lambda_j^1 - \lambda_i^1 \lambda_j^0 \tag{3.5}$$

$$[ij] = [ij] = \epsilon^{\dot{\alpha}\dot{\beta}} \tilde{\lambda}_{i\dot{\alpha}} \tilde{\lambda}_{j\dot{\beta}} \tag{3.6}$$

with $\epsilon^{01} = -\epsilon^{10} = 1$ (Levi-Civita). These are antisymmetric: $\langle ij \rangle = -\langle ji \rangle$, $[ij] = -[ji]$.

The dot product of two massless momenta is encoded by:

$$2\, p_i \cdot p_j = \mathrm{tr}(\not{p}_i \not{p}_j) = \langle ij \rangle [ji]. \tag{3.7}$$

In particular $p_i^2 = 0$ implies $\langle ii \rangle = [ii] = 0$.

3.3 Affine (Inhomogeneous) Spinor Coordinates

For explicit calculations (as in arXiv:2602.12176v1, Eq. (2)), one fixes the little-group gauge and writes the spinors in affine coordinates on $\mathbb{CP}^1$:

$$|i\rangle \equiv \lambda_i = \begin{pmatrix} 1 \\ z_i \end{pmatrix}, \qquad |i] \equiv \tilde{\lambda}_i = \omega_i \begin{pmatrix} 1 \\ \tilde{z}_i \end{pmatrix}, \tag{3.8}$$

where $z_i, \tilde{z}_i \in \mathbb{C}$ are inhomogeneous (affine) coordinates on the Riemann sphere and $\omega_i \in \mathbb{C}^*$ carries the little-group weight (so $\omega_i \to t^{-1}\omega_i$ under $\lambda \to t\lambda$). In this gauge the spinor brackets read:

$$\langle ij\rangle = z_j - z_i, \tag{3.9}$$

$$[ij] = \omega_i\omega_j\,(\tilde{z}_j - \tilde{z}_i). \tag{3.10}$$

The dot product identity (3.7) then becomes $2p_i \cdot p_j = (z_j - z_i)\,\omega_i\omega_j(\tilde{z}_i - \tilde{z}_j)$. This coordinate patch is particularly convenient for the half-collinear kinematics relevant to single-minus amplitudes (§4.4).

3.4 Schouten Identity

For any three spinors λ_i, λ_j, λ_k:

$$\langle ij\rangle\lambda_{k\alpha} + \langle jk\rangle\lambda_{i\alpha} + \langle ki\rangle\lambda_{j\alpha} = 0 \tag{3.11}$$

Equivalently:

$$\langle ij\rangle\langle kl\rangle = \langle ik\rangle\langle jl\rangle + \langle il\rangle\langle kj\rangle \tag{3.12}$$

3.5 Polarization Vectors

For a gluon of momentum p and helicity $\pm$:

$$\varepsilon^+_\mu = \frac{\langle r|\gamma_\mu|p]}{\sqrt{2}\langle rp\rangle} \tag{3.13}$$

$$\varepsilon^-_\mu = \frac{\langle p|\gamma_\mu|r]}{\sqrt{2}[pr]} \tag{3.14}$$

where r is an arbitrary reference spinor.

3.6 Momentum Twistors

For planar amplitudes, momentum twistors $\mathcal{Z}_i^A = (\lambda_i^\alpha, \mu_i^{\dot\alpha})$ provide a dual description where:

$$x_i^{\alpha\dot\alpha} - x_{i+1}^{\alpha\dot\alpha} = p_i^{\alpha\dot\alpha} = \lambda_i^\alpha\tilde{\lambda}_i^{\dot\alpha} \tag{3.15}$$

The dual conformal symmetry becomes manifest in these variables.

Chapter 4

Maximally Helicity Violating (MHV) Amplitudes

4.1 Why "Maximally Helicity Violating"?

For a tree-level n-gluon amplitude $A_n(h_1, \ldots, h_n)$ with $h_i = \pm 1$:

- Let m_- denote the number of negative-helicity gluons.
- Naive Yang–Mills vertices with all same helicity ($+\cdots+$ or $-\cdots-$) are strongly suppressed — amplitudes with $m_- = 0$ or $m_- = n$ vanish at tree level (and at all loops).
- Amplitudes with $m_- = 1$ (single-minus) were long conjectured to vanish; see §4.4.
- In twistor space (Witten 2004), tree amplitudes are supported on curves of degree $d = m_- - 1 + \ell$, so at tree level $\ell = 0$ the non-zero sector requires $0 \le m_- \le 4$ generically.

MHV = maximally helicity violating corresponds to $m_- = 2$, i.e. $m_+ = n - 2$. This is the maximum number of "wrong-helicity" gluons that gives a generically non-zero amplitude, in the sense that every cyclic ordering is non-zero. The name reflects that having only 2 negative-helicity gluons out of n is the extreme violation of a naive helicity-conservation intuition.

4.2 The Parke-Taylor Formula

For n-gluon scattering with legs j and k carrying negative helicity, the tree-level color-ordered amplitude is:

$$\boxed{A_n^{\mathrm{MHV}}(\ldots, j^-, \ldots, k^-, \ldots) = i\frac{\langle jk\rangle^4}{\langle 12\rangle\langle 23\rangle\cdots\langle n1\rangle}} \tag{4.1}$$

(momentum-conserving delta function $\delta^4(P)$ implicit). For the canonical choice $(j, k) = (1, 2)$ this reads $i\,\langle 12\rangle^4/(\langle 12\rangle\langle 23\rangle\cdots\langle n1\rangle)$.

This compact formula replaces $O(n!)$ Feynman diagrams with a single expression.

Little group weight check. Under $\lambda_i \to t_i\lambda_i$, $\tilde{\lambda}_i \to t_i^{-1}\tilde{\lambda}_i$, a helicity-$h_i$ gluon requires the amplitude to scale as $t_i^{-2h_i}$.

- Negative-helicity legs j, k: numerator $\langle jk\rangle^4 \sim t_j^4 t_k^4$, denom contributes $t_j^{-1}t_k^{-1}$ (one factor each), net $t_j^4/t_j^{-1} = t_j^{4+1}$... more carefully: $\langle jk\rangle^4$ in numerator gives weight $+4$ for j and $+4$ for k; each $\langle\cdot j\cdot\rangle$ in denominator gives -1; there are exactly 2 such factors, so net weight for j is $4 - 2 = +2 = -2h_j$ with $h_j = -1$. ✓
- Positive-helicity legs: appear twice in the denominator (once as $\langle i{,}i{+}1\rangle$ and once as $\langle i{-}1{,}i\rangle$), giving weight $-2 = -2(+1)$. ✓

4.3 Helicity Configuration Table

Helicity Configuration	Tree Amplitude	Status
$(-,-,+,+,\ldots,+)$	$\neq 0$	MHV
$(-,+,+,+,\ldots,+)$	$\neq 0$	MHV (Parke-Taylor)
$(+,-,+,+,\ldots,+)$	$\neq 0$	MHV (cyclic)
$(-,-,-,+,\ldots,+)$	$\neq 0$	NMHV
$(-,+,+,+,+,+)$	$\neq 0$	Single-minus (2026 result)
$(+,+,+,+,+,+)$	$= 0$	Vanishing

4.4 Single-Minus Amplitudes Are Nonzero

Recent work (Guevara, Lupsasca, Skinner, Strominger, Weil, arXiv:2602.12176) proves that single-minus ($m_- = 1$) tree amplitudes are non-zero for all $n \geq 4$.

$$A_n(1^-, 2^+, \ldots, n^+) \neq 0 \quad \forall\, n \geq 4. \tag{4.2}$$

This breaks the longstanding conjecture that only $m_- \geq 2$ configurations were generically nonvanishing.

Half-collinear locus (arXiv:2602.12176 Eq. (13)). The single-minus amplitude is supported on the *half-collinear locus*:

$$\langle ij \rangle = 0 \quad \forall\, i, j \in \{1, \ldots, n\}. \tag{4.3}$$

This means all λ_i are proportional, i.e. $\lambda_i = c_i\, \lambda_1$ for some $c_i \in \mathbb{C}$, so all n momenta are collinear. The $\delta(\langle 1a \rangle)$ factors in the ansatz below enforce this support explicitly for $a = 2, \ldots, n$.

Explicit single-minus ansatz (arXiv:2602.12176 Eq. (15)). The ansatz is:

$$\begin{aligned} \mathcal{A}_n(1^-, 2^+, \ldots, n^+) = i^{2-n}\, \frac{\langle r1 \rangle^{n+1}}{\langle r2 \rangle \langle r3 \rangle \cdots \langle rn \rangle}\, A_{1\cdots n} \prod_{a=2}^{n} \delta(\langle 1a \rangle) \\ \times\ \delta^2\Big(\sum_{i=1}^{n} \langle ri \rangle\, \tilde{\lambda}_i\Big) \end{aligned} \tag{4.4}$$

where r is the reference spinor, $A_{1\cdots n}$ is a scalar prefactor, and $\delta^2(\cdots)$ is a 2-component ($\dot{\alpha}$) delta function imposing a residual momentum constraint. In affine coordinates (§3.3), $\langle 1a \rangle = z_a - z_1$, so the delta functions set $z_a = z_1$ for all a.

MHV comparison (arXiv:2602.12176 Eq. (11)). For comparison, the MHV amplitude in the same notation with negative-helicity legs r, s reads:

$$\mathcal{A}_n^{\mathrm{MHV}} = i\, \langle rs \rangle^4\, \mathrm{PT}_{\mathrm{cyc}}\, \delta^4\Big(\sum_{k=1}^{n} p_k\Big), \tag{4.5}$$

where $\mathrm{PT}_{\mathrm{cyc}} \equiv 1/(\langle 12 \rangle \langle 23 \rangle \cdots \langle n1 \rangle)$ is the cyclic Parke-Taylor denominator. The full 4-momentum-conserving delta function $\delta^4(\sum_k p_k)$ is displayed explicitly here (it was left implicit in (4.5)). The structural contrast with (4.4) is striking: MHV has a standard $\delta^4(P)$ on generic kinematics, while single-minus is localized on the half-collinear locus (4.3) via the $\prod_a \delta(\langle 1a \rangle)$ factors.

4.5 BCJ Relations

The Bern-Carrasco-Johansson (BCJ) relations express amplitude redundancies:

$$\sum_{i=2}^{n-1} \frac{s_{1i}}{\langle 1i\rangle[i1]} A_n(2,\ldots,i,1,i+1,\ldots,n) = 0 \tag{4.6}$$

These reduce the number of independent color-ordered amplitudes from $(n-3)!$ to $(n-3)!/2$.

4.6 Double Copy

Gauge theory amplitudes factorize into gravity amplitudes via:

$$M_n^{\text{gravity}} = \sum_{\text{duals}} \frac{A_n^{\text{gauge}} \tilde{A}_n^{\text{gauge}}}{\text{kinematic factors}} \tag{4.7}$$

This is the KLT (Kawai-Lewellen-Tye) relation.

Chapter 5

On-Shell Recursion (BCFW)

5.1 The BCFW Shift

Choose two legs i and j. Deform their spinors:

$$\lambda_i \to \lambda_i + z\lambda_j \tag{5.1}$$

$$\tilde{\lambda}_j \to \tilde{\lambda}_j - z\tilde{\lambda}_i \tag{5.2}$$

Under this shift:

$$p_i(z) + p_j(z) = p_i + p_j \quad \text{(momentum conserved)} \tag{5.3}$$

5.2 The Recursion Formula

If $A_n(z) \to 0$ as $z \to \infty$, then by Cauchy's theorem:

$$\boxed{A_n = \sum_{\text{channels } P} A_L^{\text{on-shell}} \frac{1}{P^2} A_R^{\text{on-shell}}} \tag{5.4}$$

where the sum runs over all factorization channels where P is on-shell ($P^2 = 0$).

5.3 Why It Works

Traditional Feynman diagrams explode combinatorially:

n gluons	Feynman diagrams
4	4
5	25
6	220
10	$\sim 10^7$

BCFW recursion reduces this to $O(n^3)$ complexity.

5.4 Application to Single-Minus

The arXiv:2602.12176 paper uses BCFW with:

- All-plus reference leg
- Single-minus shifted leg
- Induction on n

This proves the nonvanishing of single-minus amplitudes for all n.

5.5 Soft Limits

As $p_n \to 0$:

$$A_n \to \frac{\langle n-1, n\rangle}{\langle n, 1\rangle} A_{n-1} \tag{5.5}$$

This provides a consistency check on amplitude formulas.

Part III

Computational Tools

Chapter 6

Computational Tools

6.1 Recommended Tool Stack

Priority order for OpenClaw physics agent:

1. **lips** — Spinor-helicity brackets, phase-space generation
2. **SymPy** — Gamma matrix algebra, symbolic manipulation
3. **CADABRA2** — Tensor calculus, field theory
4. **FORM** — Large-scale polynomial algebra

6.2 lips

```
pip install lips
```

Features:

- Spinor brackets $\langle ij \rangle$, $[ij]$
- Massless phase-space over $\mathbb{C}$, $\mathbb{F}_p$, $\mathbb{Q}_p$
- Symbol alphabet generation
- 2-loop amplitudes via `antares-results`

6.3 SymPy

```
import sympy as sp
from sympy import symbols, simplify, latex

# Spinor bracket
def angle_bracket(i, j, lam):
    return lam[i][0]*lam[j][1] - lam[i][1]*lam[j][0]
```

6.4 CADABRA2

```
sudo apt install cadabra2
```

Python bindings available. Best for tensor operations in field theory.

6.5 FORM

Industrial-scale. No Python API — subprocess only:

```
import subprocess
result = subprocess.run(['form', 'script.frm'], capture_output=True)
```

Part IV

Research Frontiers

Chapter 7

Research Frontiers

7.1 The Amplituhedron

Scattering amplitudes as volumes of positive geometries in $G(k, n)$.

7.2 Celestial Amplitudes

Amplitudes in Mellin space, 2D CFT structure, BMS symmetry.

7.3 Single-Minus Open Problems

From arXiv:2602.12176:

- Loop-level generalizations
- Soft factorization properties
- Relation to celestial amplitudes

7.4 Twistor String Theory

Degree-k curves in twistor space, CHY scattering equations.